HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F79MA

STATISTICAL MODELS A

IN-PERSON EXAMINATION

December Examination and Assessment Diet 2024-2025

Duration: 2 Hours

Attempt ALL questions.
Appropriate working must be shown to obtain full credit.

Question 1	17
Question 2	11
Question 3	15
Question 4	12
Question 5	5
Total	60

Resources Included within Exam Paper: Standard Formula sheet

Additional Resources to be Supplied to Candidates: New Cambridge Statistical Tables (Blue)

Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

DO NOT REMOVE THIS PAPER FROM THE EXAM HALL

1. (a) Let $\underline{X} = (X_1, X_2, \ldots, X_n)$ be an i.i.d. random sample from a uniform U$(0, \theta)$ distribution, having probability density function given by

$$f(x;\theta) = \frac{1}{\theta}, \quad 0 < x < \theta,$$

with $\theta > 0$ being an unknown parameter.
[**Hint**: For the uniform distribution on (0, θ), the jth order statistic $\frac{X_{(j)}}{\theta} \sim \text{Beta}(j, n+1-j)$]

i. Estimate θ using the method of moments (MME) and maximum likelihood estimation (MLE), denoting the estimators as $\tilde{\theta}$ and $\hat{\theta}$, respectively.

(3 marks)

ii. Find the mean and variance of each of the MME estimator $\tilde{\theta}$ and the MLE estimator $\hat{\theta}$. Are both estimators unbiased?

(5 marks)

iii. Derive the mean square errors (MSE) of each of the MME estimator $\tilde{\theta}$ and the MLE estimator $\hat{\theta}$. Which estimator should be preferred? Explain.

(3 marks)

iv. Determine an expression for a confidence interval for θ using the MME $\tilde{\theta} = 2\bar{X}$, assuming a Normal approximation to the distribution of $\tilde{\theta}$.

(3 marks)

(b) Suppose that $\underline{X} = (X_1, X_2, \ldots, X_n)$ is an i.i.d. random sample from a Beta distribution with parameters α and β, where α is known and β is unknown ($\beta > 0$).
The probability density function of the Beta distribution is given by:

$$f(x;\alpha,\beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, \quad 0 < x < 1,$$

where $\Gamma(\cdot)$ represents the Gamma function.
Show that

$$T = \frac{\left(\sum_{i=1}^{n} \log(1-X_i)\right)^3}{n}$$

is a sufficient statistic for the parameter β.

(3 marks)
[Total 17 marks]

2. Let $\underline{X} = (X_1, X_2, \ldots, X_n)$ be an i.i.d. random sample from a distribution having probability density function given by

$$f(x;\theta) = \frac{\theta^{1/3}}{\Gamma(4/3)} \exp(-\theta x^3), \quad 0 < x < \infty,$$

where $\theta > 0$ is an unknown parameter and $\Gamma(z)$, as usual, denotes the Gamma function given by

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt.$$

(a) Show that the score function $U(\theta)$ is given by

$$U(\theta) = \frac{n}{3\theta} - \sum_{i=1}^{n} X_i^3.$$

(3 marks)

(b) Find the maximum likelihood estimator $\hat{\theta}$ of θ.

(1 mark)

(c) Without doing any calculations, is $\hat{\theta}$ unbiased? Give a reason for your answer.

(2 marks)

(d) Calculate the Fisher information $I(\theta)$ and hence write down the asymptotic distribution of $\hat{\theta}$.

(2 marks)

(e) Suppose you are given a sample $\underline{x} = (x_1, \ldots, x_{15})$ of size 15 such that

$$\sum_{i=1}^{15} x_i^3 = 5.$$

Calculate $\hat{\theta}$ for this sample and hence show that with this estimated value of θ we have

$$\mathbf{E}(X^3) = \frac{\sum_{i=1}^{n} x_i^3}{n}.$$

Hint: You may find it helpful to consider the definition of $\Gamma(4/3)$

(3 marks)

[Total 11 marks]

3. Let $X \sim N(0, \sigma^2)$, and consider testing $H_0 : \sigma = \sigma_0$ versus $H_1 : \sigma = \sigma_1$, where $\sigma_1 > \sigma_0$. The values σ_0 and σ_1 are fixed.

   (a) Find an expression for the likelihood ratio as a function of x. Hence, identify the values of x that favour H_0 and specify a rejection region for a level α test.

   (4 marks)

   (b) For an i.i.d. random sample of $X_1, X_2, \ldots, X_n$ distributed as $N(0, \sigma^2)$, derive the likelihood ratio and the rejection region for the same hypotheses as in part (a).

   (4 marks)

   (c) Is the likelihood ratio test derived in part (b) uniformly most powerful for testing $H_0 : \sigma = \sigma_0$ against $H_1 : \sigma > \sigma_0$?

   (4 marks)

   (d) Perform numerical calculations for the following scenario: Consider a sample of size $n = 10$, with $\sigma_0 = 1$, $\sigma_1 = 1.5$, and an observed value $\sum_{i=1}^{10} x_i^2 = 14$. At a significance level of $\alpha = 0.05$, does the test reject H_0?

   (3 marks)

[Total 15 marks]

4. Suppose that you work as an actuary at an insurance company. As an actuary, you are interested in understanding the arrival times of insurance claims at the company. Denote by X the arrival time of an insurance claim. You are told that the times at which the insurance claims arrive at the company follow a continuous distribution with the following density function:

$$f_X(x) = 3\theta x^2 e^{-\theta x^3}, \text{ for } x > 0,$$

where $\theta > 0$ is a rate parameter.

Suppose that you have observed times $x_1, x_2, \cdots, x_n$ for n insurance claims throughout the year. To perform Bayesian analyses, you decide to use the prior distribution $\theta \sim \text{Gamma}(\alpha, \beta)$. Answer the following questions:

(a) Show that the posterior distribution of $(\theta|\underline{\text{x}})$ also follows the gamma distribution. State clearly the parameters of the distribution of $(\theta|\underline{\text{x}})$. What is the Bayes estimate for θ under the minimisation of the mean square error?

(4 marks)

(b) Express the Bayes estimate found in part (a) in the form of $ZT + (1 - Z)\mu$, where $0 \leq Z \leq 1$ is a weighting factor, T is a statistic of $\underline{\text{x}}$ (or a transformation of the statistic), and μ is a component from our prior information. State clearly the values of Z, T, and μ. Also, describe the statistic T in words.

(5 marks)

(c) As we collect more samples, i.e., as n grows larger, explain what would happen to the values of

i. the weighting factor Z,

ii. the component μ,

iii. the Bayes estimate for θ, $\mathbb{E}(\theta|\underline{\text{x}})$,

Justify your answers.

(3 marks)

[Total 12 marks]

5. The table below shows the aggregate claims in five successive years from 4 insurance policies. The monetary amounts have been adjusted to account for the effects of inflation, and they are given in units of £1000.

Policy number (i)	Year (j) 1	2	3	4	5	$\bar{X}_i = \frac{1}{n}\sum_{j=1}^{n} X_{ij}$	$S_i^2 = \frac{1}{n-1}\sum_{j=1}^{n}(X_{ij} - \bar{X}_i)^2$
1	70	85	73	91	88	81.4	87.3
2	95	90	82	100	91	91.6	44.3
3	61	42	50	67	37	51.4	158.3
4	54	51	53	60	57	55	12.5

Using the Empirical Bayes Credibility Theory Model 1 (Bühlmann model), calculate the credibility premium for the coming years for policy number 4.

[Total 5 marks]

END OF PAPER (TOTAL 60 marks)

Some Continuous Distributions

Name	Notation	p.d.f.	Range	m.g.f.	Mean	Variance
Uniform	$\mathrm{U}(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{e^{bt}-e^{at}}{(b-a)t}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$\mathrm{N}(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$-\infty < x < \infty$	$e^{\mu t+\frac{1}{2}\sigma^2 t^2}$	μ	σ^2
Log Normal	$\mathrm{LogN}(\mu,\sigma^2)$	$\frac{1}{x\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(\ln x-\mu)^2}{2\sigma^2}\right\}$	$0 < x < \infty$	undefined for $t > 0$	$e^{\mu+\frac{1}{2}\sigma^2}$	$(e^{\sigma^2}-1)(e^{2\mu+\sigma^2})$
Gamma	$\mathrm{Gamma}(\alpha,\lambda)$	$\frac{\lambda^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\lambda x}$	$0<x<\infty$, $\alpha,\lambda>0$	$\left(\frac{\lambda}{\lambda-t}\right)^\alpha$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
Exponential	$\mathrm{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0<x<\infty$, $\lambda>0$	$\frac{\lambda}{\lambda-t}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Chi-square	χ^2_ν	$\frac{1}{2^{\nu/2}\Gamma(\nu/2)}x^{\nu/2-1}e^{-x/2}$	$0<x<\infty$, $\nu>0$ integer	$(1-2t)^{-\nu/2}$	ν	2ν
Beta	$\mathrm{Beta}(\alpha,\beta)$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$	$0<x<1$, $\alpha,\beta>0$	*omitted	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Cauchy	$\mathrm{Cauchy}(\mu,\sigma)$	$\frac{\sigma}{\pi(\sigma^2+(x-\mu)^2)}$	$\sigma > 0$	does not exist		
Pareto	$\mathrm{Pareto}(k,\theta)$	$\frac{k}{\theta}\left(1+\frac{x}{\theta}\right)^{-(k+1)}$	$x>0$, $\theta,k>0$	does not exist	$\frac{\theta}{k-1}$	$\frac{\theta^2 k}{(k-2)(k-1)^2}$
Weibull	$\mathrm{Weibull}(\nu,\lambda)$	$\nu\lambda x^{\nu-1}\exp(-\lambda x^\nu)$	$x>0$, $\nu,\lambda>0$	*omitted	$\lambda^{-\frac{1}{\nu}}\Gamma(1+\frac{1}{\nu})$	*omitted
Multivariate Normal	$\mathrm{N}_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$	$\frac{\exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}}{(2\pi)^{p/2}\det(\boldsymbol{\Sigma})^{1/2}}$	$\mathbf{x}\in\mathbb{R}^p$	$e^{\boldsymbol{\mu}^{\mathrm{T}}\mathbf{t}+\frac{1}{2}\mathbf{t}^{\mathrm{T}}\boldsymbol{\Sigma}\mathbf{t}}$	$\boldsymbol{\mu}$	covariance matrix $\boldsymbol{\Sigma}$

Some Discrete Distributions

Name	Notation	p.m.f.	Range	m.g.f.	Mean	Variance
Discrete Uniform	$\mathrm{DU}(a,b)$	$\frac{1}{b-a+1}$	$x = a,a+1,\ldots,b-1,b$ a,b integers	$\frac{e^{at}-e^{(b+1)t}}{(b-a+1)(1-e^t)}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$
Binomial	$\mathrm{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$x = 0,1,2,\ldots,n$ $0<p<1$	$[(1-p)+pe^t]^n$	np	$np(1-p)$
Poisson	$\mathrm{Poi}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$x = 0,1,2,\ldots$ $\lambda>0$ real	$e^{\lambda(e^t-1)}$	λ	λ
Hyper-geometric	$\mathrm{HypGeo}(n,K,N)$	$\frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$	$x = 0,1,2,\ldots,n$ N,K,n integers	*omitted	$\frac{nK}{N}$	$\frac{nK(N-K)(N-n)}{(N)^2(N-1)}$
Geometric	$\mathrm{Geo}(p)$	$p(1-p)^{x-1}$	$x = 1,2,\ldots$ $0<p<1$	$p[e^{-t}-(1-p)]^{-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative Binomial	$\mathrm{NegBin}(k,p)$	$\binom{x-1}{k-1}p^k(1-p)^{x-k}$	$x = k,k+1,\ldots$ $0<p<1$	$p^k[e^{-t}-(1-p)]^{-k}$	$\frac{k}{p}$	$\frac{k(1-p)}{p^2}$

*omitted due to lack of space

Empirical Bayesian Credibility theory Model 1 (Buhlmann Model)

Credibility factor

$$Z = \frac{n}{n + \frac{\widehat{\mathbf{E}[s^2(\theta_i)]}}{\widehat{\mathbf{Var}[m(\theta_i)]}}}$$

with the estimates

$$\widehat{\mathbf{E}[m(\theta_i)]} = \frac{1}{N}\sum_{i=1}^{N} \bar{X}_i,$$

$$\widehat{\mathbf{E}[s^2(\theta_i)]} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1}{n-1}\sum_{j=1}^{n}\left(X_{ij} - \bar{X}_i\right)^2\right],$$

and

$$\widehat{\mathbf{Var}[m(\theta_i)]} = \frac{1}{N-1}\sum_{i=1}^{N}\left(\bar{X}_i - \bar{X}\right)^2 - \frac{1}{nN}\sum_{i=1}^{N}\left[\frac{1}{n-1}\sum_{j=1}^{n}\left(X_{ij} - \bar{X}_i\right)^2\right].$$